

**Table 5.4 – Recommended maximum length of operation under shuttle flow**

| Traffic volume (Total for both directions) (vph) | Length of single lane section (m) |
|--------------------------------------------------|-----------------------------------|
| 701–800                                          | 70                                |
| 601–700                                          | 100                               |
| 501–600                                          | 150                               |
| 401–500                                          | 250                               |
| 351–400                                          | 400                               |
| 301–350                                          | 600                               |
| ≤ 300                                            | 800                               |

The volumes in Table 5.4 have been determined to allow a quick analysis without referring to a traffic engineering professional. Additional traffic engineering input is required to support a longer length of single-lane operation.

Generally, when using Table 5.4, and where the lengths are within the maximum limits, single-lane operation using active control by portable traffic control devices or traffic controllers will lead to a relatively short and consistent or stable queue length; however, additional traffic engineering input and considerations (risk assessments) are required to support longer lengths of single-lane operation which will generally lead to longer maximum queue lengths and queue lengths that are not easily managed, or are variable and unstable. Contingency planning for longer than expected or continually growing queue lengths shall be included as part of the TGS design. End-of-queue protection measures (see Section 4.8.3) shall be considered.

Figure 5.4.4(a) does not include all traffic control devices required and is not to be used as a TGS diagram.